

QC-SOP-009 — Laboratory method selection and execution

Revision D0.2 | DRAFT FOR REVIEW | Revised 2026-10-04

Effective date: not recorded. Approval/signature: not recorded. This draft does not authorize testing, release, production changes or UAT acceptance.

1. Purpose and scope

Plan and execute laboratory work using approved methods, run context, calibrated equipment and typed observations.

Implementation: SOURCE-ALIGNED DRAFT

2. Responsibilities and review

Responsible specialists: Laboratory operator; Technical method owner; Supervisor; QCM

Review: Laboratory operator; Technical method owner; Supervisor; QCM

Approval: Authorized QMS / Document Control approver under the approved document policy; not yet established for this draft

3. Before you start

- The exact method, product applicability and scientific source are approved.
- The equipment and calibration match the method and are eligible.
- The operator has the lab permissions for creation and measurement entry.

4. Procedure

1. Identify the request, receiving source where applicable, product, lot and sample identifiers. Select an approved template whose version and controlled-source context match the task.
2. Confirm NUMERIC, TEXT or BOOLEAN datatype, units, required points, precision, rounding and calculation rule for each parameter. Do not infer a datatype from a value that looks like true or false.
3. Plan the run and record its equipment and calibration context. Equipment must be ACTIVE, not under maintenance and have matching CURRENT, non-overdue calibration.
4. Prepare and verify the instrument using its approved manufacturer manual and method. Record reference checks, environment, fixture and program revision; the exact device manual is still a required source for the photographed leak-test equipment.
5. Acquire raw readings against the correct sample and run. Retain replicates, timestamps and printouts rather than keeping only a calculated average.
6. Enter observations without premature rounding or unit conversion. NUMERIC values are decimal strings, TEXT stays text and BOOLEAN is a real boolean. Remarks require an associated value.
7. Apply only an approved calculation and evaluation rule. The current production evaluator denies missing scientific authority; preserve raw work and seek review rather than manufacture a PASS.
8. Submit for the two-stage review, resolve any return and preserve version-bound final evidence. Deliver the reviewed result to the requesting process; result approval does not establish lot release.

5. Decisions, errors and stop conditions

- A missing method, calibration or calculation source prevents controlled evaluation.

- Instrument failure or ambiguous sample identity requires an invalid-run investigation and retest decision, not silent deletion.

6. Records and handoff

- Test and template snapshot
- Run and sample context
- Equipment/calibration evidence
- Raw and calculated values
- Review chain

Before handoff, reopen the saved record where the application supports it and compare the result with the source. Retain an unresolved outcome as unresolved; do not infer approval from a name, signature image, task completion or test result.

7. Sources and related documents

- src/modules/laboratory/domain/measurement.ts
- src/modules/laboratory/application/save-measurements.ts
- src/modules/laboratory/domain/lab-state.ts

Related work instructions: QC-WI-015, QC-WI-016, QC-WI-017, QC-WI-050, QC-WI-051

Repository sources are bound to HEAD 435f9f704049dee4916223b24b96866d7fa9af2e. Resolve source IDs, file hashes, external editions and form originals in References (source metadata is inside Authoring_Sources.zip).

8. Required decisions before approval

- DEC-001 — Scientific source and applicability
- DEC-002 — Leak-test equipment manual and safety
- DEC-005 — Per-point and overall evaluation
- DEC-006 — Leak calculation validation
- DEC-026 — Retest, invalid work and lab rejection

Document Control records real reviewer outcomes and actual approval/effectivity only after the controlled process. Completed proposals for these decisions are provided in References/Policy_and_Method_Decision_Book.pdf. Required source evidence and owner adoption remain explicitly recorded there.

9. Current route references

- /laboratory
- /laboratory/tests
- /laboratory/tests/new
- /laboratory/tests/[labTestId]
- /laboratory/tests/[labTestId]/execute
- /laboratory/tests/[labTestId]/review

- /laboratory/tests/[labTestId]/retests/new

Route placeholders identify existing records. Route coverage is documentation mapping, not proof that every role/state was accepted in a browser. Exact extracted field and permission strings are in [References/route-coverage.pdf](#) and [References/source-contract-register.pdf](#).

10. Detailed operating policy and review checks

These are D0.2 proposed operating controls for adoption. Official company records remain valid physical source records; this revision does not retrospectively change their outcomes. Use the operational record named below to capture execution and exceptions.

DEC-001 — Source adoption and applicability

The technical owner prepares a product matrix by item code, size, material, connector, sterile/bulk condition and intended use. For each row, identify the current company WI, product specification and external standard where applicable. Document Control compares the controlled master with the photographed copy. A newer standard edition triggers an impact review; it does not silently replace the company method.

Decision and exception check: An applicable source package passes review when the exact revision can be retrieved, every mandatory point has a source and method, and the reviewer has resolved conflicting instructions. Preserve existing physical record outcomes as recorded. Registering them digitally does not create a new approval or retrospective release.

Record: QC-REC-001 source adoption record; company form register; clause-to-point matrix.

DEC-002 — Equipment readiness and safe operation

At the instrument, read the nameplate and record asset ID, manufacturer/model, serial, controller/software version, pressure sensor ranges, fixture identity, regulator and reference device. Match the calibration certificate to that serial and range. Attach the actual operator manual and identify its sections for connection, pressurization, isolation, emergency stop and safe venting. Record pre-use checks with a known reference and fixture/background leak check under that manual.

Decision and exception check: Stop for an unreadable label, expired or mismatched calibration, damaged connection, failed reference check or an uncontrolled program. Make the system safe using the instrument procedure; retain the failed run and identify all affected work since the last acceptable check. A maintenance completion requires a documented return-to-use check.

Record: QC-REC-002 equipment readiness and run setup; equipment/calibration/maintenance records.

DEC-005 — Point evaluation and overall result

For a valid numeric run, retain the unrounded observation and compare by the exact approved operator. For visual checks, record a concrete observation and the form legend. The physical legend distinguishes acceptable, not acceptable, read remarks and not applicable. Preserve R as a review requirement and NA with a point-specific reason; neither becomes a automatic PASS.

Decision and exception check: Proposed aggregation: all required applicable points must have valid accepted outcomes before an overall pass is proposed. Any failed required point prevents a pass; missing/invalid/conflicted evidence remains HOLD/further evaluation. A reviewer resolves contradictory remarks even when an Accepted box is ticked. Lot release is a separate disposition.

Record: QC-REC-005 acceptance-point specification and review; original per-point result and remarks.

DEC-006 — Calculation validation and discrepancy handling

For an engineering check under constant effective volume and approximately isothermal conditions, propose $q = V \times \text{abs}(P_{\text{start}} - P_{\text{end}}) / \text{elapsed time}$. Use Pa, cm³ and seconds consistently; kPa converts to Pa by 1000

and litres to cm^3 by 1000. Document whether pressures are gauge or absolute and whether compensation, reference volume, compliance, background or standard-condition normalization applies. Validate against the actual manufacturer algorithm before official use.

Decision and exception check: Observed discrepancy for investigation: IMG-048 print shows $V=2.04 \text{ cm}^3$, 330.781 and 327.852 kPa, 20 s and an output 2981 $\text{Pa}\cdot\text{cm}^3/\text{s}$. The simple dimensional check gives 298.758 $\text{Pa}\cdot\text{cm}^3/\text{s}$. These differ by about tenfold. This is not proof of an incorrect machine; printed volume/time semantics or internal scaling may differ. Require an explanation and validation dataset.

Record: QC-REC-006 calculation validation: raw readings, units, formula/version, instrument output, independent result, difference and reviewer disposition.

DEC-026 — Invalid test, retest and rejection

Contain affected material, identify all runs since the last valid control and record the reason for invalidation or retest. The method owner approves sample selection, repeat count, conditions and the rule for combining or separately reporting results before retesting. Link the new run to the original and preserve the operator, time, equipment and source revisions.

Decision and exception check: A confirmed product failure proceeds to the authorized nonconformance/disposition process. A passing retest does not erase a failing result. Ambiguous results remain HOLD/further evaluation until reviewed. The current digital lab reject path remains policy-blocked where PD-38 authority is not established; no alternate route is proposed.

Record: QC-REC-019 invalidation and retest authorization; QC-SOP-012; QC-WI-020/042.

11. Official physical record workflow

Use a separate run row for each sample/replicate. Record connector type, sample count, lot and sampling area, then equipment identity/medium, assembly force, torque, time and angle. Record actual temperature and humidity in the environmental block; an empty block remains missing information.

Enter set pressure (sp), applied pressure (tp), volume (v), pressure difference, holding time and instrument leakage result in their printed units. Retain the printer ticket with serial/run identifier and timestamp. Match each ticket to its sample row; do not assume the top ticket describes every row underneath. Distinguish setpoint from achieved pressure.

F-8-2-6-05 Rev04 prints 300–330 kPa, 15–20 s and $<5000 \text{ Pa}\cdot\text{cm}^3/\text{s}$. F-8-2-6-07 Rev4 prints –80 to –88 kPa with the same time/limit. These are observed company-form values; adoption for a product requires the form/method reconciliation in DEC-004. A pressure outside the source range or a failed stability check needs investigation even if reported leakage is below the limit.

Mark the run outcome only after validity and criterion checks. Complete the overall PASS, FAIL or HOLD/FURTHER EVALUATION block separately from individual rows. Retain Tested by, Reviewed by and Approved by sections with actual identities/dates; do not fill a blank historical signature. Attach an investigation when independent calculation and printer output disagree.

Revision record

D0.2, 2026-10-04: expanded execution, exception and handoff controls; incorporated owner-confirmed official physical forms and completed technical decision proposals. Previous D0.1 sources are retained in History/D0.1 within Authoring_Sources.zip. Review and approval remain unrecorded.